

## CONROE-NORTH HOUSTON REGIONAL, TX, USA

WMO: 722444

Lat: 30.357N

Lon: 95.414W

Elev: 75

StdP: 100.43

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53902

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-2.5 | (c)<br>-1.1 | (d)<br>-8.9                   | (e)<br>1.8 | (f)<br>5.2 | (g)<br>-6.6 | (h)<br>2.2 | (i)<br>6.3 | (j)<br>9.5            | (k)<br>14.9 | (l)<br>8.7 | (m)<br>15.9 | (n)<br>1.0            | (o)<br>340 | (p)<br>0.381 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>11.8            | (c)<br>36.6     | (d)<br>24.1 | (e)<br>35.4 | (f)<br>24.4 | (g)<br>34.2 | (h)<br>24.5 | (i)<br>26.5         | (j)<br>31.1 | (k)<br>26.1 | (l)<br>30.9 | (m)<br>25.7 | (n)<br>30.6 | (o)<br>3.3           | (p)<br>180 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.5                            | 20.9 | 28.0 | 25.0 | 20.3 | 27.7 | 24.5 | 19.7 | 27.5 | 82.8          | 30.9 | 81.2 | 30.8 | 79.6 | 30.9 | 29.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.4               | 7.5  | 6.8 | -5.7                       | 38.3 | 2.1                | 1.7 | -7.2                                               | 39.5 | -8.5       | 40.5 | -9.7       | 41.4 | -11.2      | 42.6 |
|                   |      |     | -6.4                       | 27.4 | 1.9                | 0.7 | -7.8                                               | 27.9 | -8.9       | 28.3 | -10.0      | 28.7 | -11.3      | 29.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 20.0   | 10.4  | 12.7  | 16.3  | 20.0  | 23.8  | 27.1  | 28.1  | 28.5  | 25.8  | 20.6  | 15.2  | 11.2  |  |  |
|                                                                        | DBStd       | 7.60   | 5.37  | 5.61  | 4.89  | 3.94  | 3.13  | 1.77  | 1.58  | 1.78  | 2.68  | 4.28  | 5.13  | 5.25  |  |  |
|                                                                        | HDD10.0     | 162    | 61    | 33    | 8     | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 47    |  |  |
|                                                                        | HDD18.3     | 933    | 254   | 172   | 99    | 28    | 3     | 0     | 0     | 0     | 1     | 27    | 120   | 229   |  |  |
|                                                                        | CDD10.0     | 3813   | 72    | 107   | 202   | 300   | 429   | 512   | 561   | 573   | 473   | 329   | 169   | 85    |  |  |
|                                                                        | CDD18.3     | 1543   | 7     | 13    | 34    | 78    | 173   | 262   | 303   | 315   | 223   | 98    | 27    | 9     |  |  |
|                                                                        | CDH23.3     | 15437  | 22    | 60    | 213   | 607   | 1601  | 2736  | 3377  | 3579  | 2139  | 900   | 172   | 33    |  |  |
|                                                                        | CDH26.7     | 6994   | 0     | 5     | 27    | 137   | 598   | 1312  | 1686  | 1901  | 1001  | 301   | 26    | 1     |  |  |
| (14) Wind                                                              | WSAvg       | 2.5    | 2.7   | 3.0   | 3.1   | 3.2   | 2.9   | 2.3   | 2.1   | 2.0   | 1.8   | 2.1   | 2.4   | 2.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1194   | 101   | 79    | 84    | 79    | 104   | 131   | 83    | 72    | 92    | 130   | 115   | 102   |  |  |
|                                                                        | PrecMax     | 1697   | 261   | 158   | 206   | 261   | 283   | 420   | 271   | 201   | 182   | 541   | 364   | 223   |  |  |
|                                                                        | PrecMin     | 653    | 22    | 13    | 6     | 4     | 0     | 15    | 11    | 0     | 26    | 3     | 12    | 28    |  |  |
|                                                                        | PrecStd     | 234    | 59    | 47    | 46    | 58    | 78    | 87    | 59    | 56    | 46    | 127   | 75    | 45    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 25.7  | 27.4  | 28.9  | 31.4  | 34.1  | 37.0  | 37.3  | 38.5  | 37.6  | 32.8  | 29.0  | 26.2  |  |  |
|                                                                        |             | MCWB   | 18.6  | 19.4  | 20.1  | 22.0  | 23.6  | 23.8  | 24.2  | 23.6  | 22.7  | 23.5  | 22.5  | 20.7  |  |  |
|                                                                        | 2%          | DB     | 23.6  | 25.2  | 27.3  | 29.6  | 32.5  | 35.1  | 35.9  | 36.9  | 34.8  | 31.6  | 27.3  | 24.0  |  |  |
|                                                                        |             | MCWB   | 17.9  | 18.9  | 19.8  | 21.1  | 23.4  | 24.1  | 24.4  | 24.2  | 23.5  | 23.2  | 20.8  | 19.5  |  |  |
|                                                                        | 5%          | DB     | 22.0  | 23.3  | 26.0  | 28.0  | 31.4  | 33.8  | 34.7  | 35.7  | 33.2  | 30.0  | 25.4  | 22.4  |  |  |
|                                                                        |             | MCWB   | 17.7  | 18.3  | 19.1  | 20.8  | 23.3  | 24.4  | 24.6  | 24.6  | 23.3  | 22.1  | 19.5  | 19.1  |  |  |
|                                                                        | 10%         | DB     | 20.1  | 21.6  | 24.0  | 27.1  | 30.2  | 32.7  | 33.6  | 34.3  | 32.1  | 28.1  | 23.6  | 21.0  |  |  |
|                                                                        |             | MCWB   | 16.6  | 17.9  | 18.1  | 20.2  | 23.0  | 24.4  | 24.8  | 24.7  | 23.4  | 21.1  | 19.4  | 18.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 20.9  | 21.6  | 22.4  | 24.3  | 26.0  | 26.9  | 26.9  | 26.9  | 26.4  | 25.8  | 23.9  | 22.1  |  |  |
|                                                                        |             | MCDB   | 23.1  | 23.8  | 26.3  | 28.1  | 31.0  | 31.1  | 31.1  | 31.8  | 30.4  | 29.2  | 26.6  | 24.0  |  |  |
|                                                                        | 2%          | WB     | 19.7  | 20.7  | 21.5  | 23.3  | 25.2  | 26.2  | 26.4  | 26.4  | 25.8  | 25.1  | 22.5  | 21.2  |  |  |
|                                                                        |             | MCDB   | 21.8  | 23.4  | 24.7  | 27.1  | 29.9  | 30.8  | 31.1  | 31.5  | 30.0  | 28.8  | 25.1  | 23.1  |  |  |
|                                                                        | 5%          | WB     | 18.9  | 19.8  | 20.8  | 22.5  | 24.6  | 25.7  | 25.9  | 26.1  | 25.4  | 24.2  | 21.6  | 20.2  |  |  |
|                                                                        |             | MCDB   | 21.3  | 22.2  | 23.5  | 26.0  | 28.9  | 30.4  | 30.7  | 31.2  | 29.7  | 27.5  | 24.1  | 22.1  |  |  |
|                                                                        | 10%         | WB     | 17.3  | 18.7  | 20.0  | 21.8  | 24.0  | 25.2  | 25.6  | 25.7  | 24.8  | 23.3  | 20.6  | 18.6  |  |  |
|                                                                        |             | MCDB   | 19.5  | 21.0  | 22.7  | 25.1  | 28.0  | 30.1  | 30.5  | 30.9  | 29.1  | 26.3  | 22.9  | 20.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.2  | 11.8  | 12.4  | 12.2  | 11.5  | 11.0  | 10.9  | 11.8  | 11.9  | 13.2  | 12.7  | 12.0  |  |  |
|                                                                        |             | MCDBR  | 13.4  | 12.7  | 12.5  | 12.6  | 12.2  | 12.8  | 12.5  | 13.5  | 13.7  | 13.5  | 13.1  | 13.0  |  |  |
|                                                                        |             | MCWBR  | 8.5   | 7.4   | 6.1   | 5.7   | 4.7   | 4.1   | 3.6   | 3.7   | 4.6   | 5.9   | 7.3   | 8.9   |  |  |
|                                                                        | 5% WB       | MCDBR  | 10.5  | 10.2  | 9.8   | 9.7   | 10.0  | 10.5  | 10.8  | 11.6  | 10.5  | 10.9  | 10.3  | 10.8  |  |  |
|                                                                        |             | MCWBR  | 7.8   | 7.0   | 5.9   | 5.1   | 4.3   | 4.0   | 3.8   | 3.8   | 4.0   | 5.5   | 6.6   | 8.6   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.332 | 0.356 | 0.371 | 0.427 | 0.476 | 0.466 | 0.490 | 0.475 | 0.446 | 0.376 | 0.370 | 0.339 |  |  |
|                                                                        | taud        |        | 2.514 | 2.460 | 2.411 | 2.257 | 2.157 | 2.237 | 2.197 | 2.239 | 2.281 | 2.462 | 2.425 | 2.501 |  |  |
|                                                                        | Ebn at Noon |        | 901   | 907   | 913   | 869   | 825   | 829   | 807   | 816   | 828   | 873   | 847   | 872   |  |  |
|                                                                        | Edh at Noon |        | 88    | 102   | 114   | 138   | 153   | 141   | 146   | 139   | 128   | 100   | 96    | 86    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.86  | 3.19  | 4.38  | 5.31  | 5.83  | 6.15  | 6.17  | 5.88  | 5.03  | 4.26  | 3.19  | 2.50  |  |  |
|                                                                        | RadStd      |        | 0.27  | 0.59  | 0.40  | 0.34  | 0.44  | 0.50  | 0.50  | 0.44  | 0.48  | 0.55  | 0.33  | 0.31  |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (36 neighbors) | +0.34 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +110    | +95     |

Nomenclature: See separate page